

Eduardo Gusmao

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SUMMARY

Data Platform Engineer with 8+ years of experience designing and operating scalable data platforms, distributed data systems, and production-grade pipelines across healthcare, life sciences, and enterprise environments. Strong expertise in Python and SQL, with hands-on experience in PySpark/Spark, Kafka-style event-driven workflows, Airflow, AWS, Docker/Kubernetes, and CI/CD to build reliable data infrastructure and platform services.

Proven track record delivering end-to-end data platform solutions, including ingestion frameworks, transformation layers, validation, data modeling, observability, and internal tooling, with strong focus on reliability, performance, data quality, and operational excellence. Experienced in batch and streaming-adjacent pipelines, distributed processing, cloud/on-prem patterns, and modern formats such as Parquet and JSON.

Collaborative and impact-driven, working closely with analysts, data scientists, engineers, product, and infrastructure teams to translate complex requirements into scalable, maintainable platform capabilities that support analytics, BI, ML workflows, and mission-critical data products.

PROFESSIONAL EXPERIENCE

Senior Data Engineer

Turku Biosciences / Brazilian Ministry of Health | Finland / Brazil | Sep 2022 - Present

- Lead the design and delivery of a national-scale data platform for precision medicine, integrating multi-modal genomic, phenotypic, and clinical (EHR) data for 65,000+ individuals through scalable Python-based pipelines, distributed systems (PySpark/Spark, Databricks, HPC/SLURM), and API-driven ingestion workflows.
- Architected and deployed production-grade data platform services for regulated ingestion, validation, and transformation, implementing data governance, PII-compliant processing, reproducible workflows, and robust quality/reliability controls across the full lifecycle (monitoring, idempotency, consistency).
- Developed and optimized high-performance processing and modeling layers using Python, SQL, and PySpark, improving pipeline efficiency (~25%) and reducing storage costs (~80%) through partitioning strategies, optimized formats (Parquet), and distributed query tuning.
- Designed reusable integration layers and backend data services connecting heterogeneous sources (clinical, genomic, ERP/SAP), enabling event-driven workflows, orchestration patterns, and batch/streaming-adjacent pipelines supporting analytics, BI, and ML systems with >40% faster data delivery.
- Collaborated with product, analytics, engineering, and infrastructure stakeholders to deliver scalable platform capabilities, CI/CD, Docker+K8s workloads, observability (logging, alerting), and performance tuning across DE.

Data Engineer II

Göttingen General Hospital | Germany | Mar 2019 - Sep 2022

- Designed and implemented scalable data platform services using Python and SQL on cloud and hybrid environments, enabling reliable ingestion, transformation, and low-latency access for downstream analytics, BI, and apps workloads.
- Developed and optimized high-performance ETL/ELT pipelines with Python and PySpark, leveraging distributed processing, batch workflows, and orchestration patterns to improve data availability and system responsiveness (~33%).
- Refactored legacy systems into modular, production-grade platform services, introducing CI/CD pipelines, automated testing, monitoring/logging, and robust error handling (idempotency, retries), significantly improving reliability, maintainability, and operational efficiency (~50-60%).
- Built reusable data processing frameworks and integration layers for large-scale, heterogeneous datasets, applying data modeling, validation, lifecycle standards, and governance across 12 cross-functional teams in a distributed environment.

Data Engineer I

Dana-Farber Cancer Institute | USA | Jan 2016 - Mar 2019

- Developed cloud-native data platform services supporting large-scale drug discovery, building Python-based ETL/ELT pipelines and API-driven integration layers for heterogeneous biomedical, operational, and financial datasets.
- Implemented end-to-end data processing pipelines using Python, SQL, and PySpark on distributed systems (Apache Spark), enabling scalable ingestion, transformation, and batch workflows for analytics and ML-driven applications.
- Collaborated with cross-functional stakeholders (product, analytics, research) to define KPIs and translate requirements into efficient data models, backend data logic, and reusable platform components, improving data accessibility and reducing operational costs by >25%.
- Contributed to production-grade data engineering practices, including Git version control, validation checks, structured documentation, and maintainable system design, ensuring reliability and reproducibility across the data lifecycle.

EDUCATION

Ph.D. in Biomedical Informatics

Harvard Medical School | Boston/Cambridge, USA | 2013 - 2017

Ph.D. (Dr. rer. nat.) in Data Engineering & Computational Life Sciences

RWTH Aachen University | Aachen, Germany | 2011 - 2015

B.Sc. & M.Sc. in Computer Science

Federal University of Pernambuco (UFPE) | Recife, Brazil | 2008 - 2011

SKILLS

Languages

English (Fluent); Portuguese (Native); Spanish (B2); German (A2); Finnish (A1).

Programming

Python (advanced; data platforms, pipelines, APIs, automation); SQL (advanced); Bash/Shell; working knowledge of Scala, Java, C/C++; REST APIs, async programming, data serialization (Parquet, JSON), and production-grade software engineering.

Data Platform Engineering

Design and operation of scalable data platforms, distributed systems, and internal data tooling; reusable ingestion frameworks, transformation layers, platform APIs, developer-facing abstractions, and self-service data capabilities supporting analytics, ML, and BI workloads.

Distributed Data Systems

PySpark, Spark, Pandas, Polars, NumPy; large-scale processing, distributed compute, performance tuning, partitioning, query optimization, and resource efficiency; experience with batch and streaming-adjacent pipelines, Kafka concepts, and Spark Streaming patterns.

Cloud, Hybrid & Infrastructure

AWS, Azure, GCP; cloud-native and hybrid data infrastructure; HPC/SLURM; Docker, Kubernetes, Terraform; infrastructure-aware engineering, containerized workloads, deployment environments, and scalable platform operations.

Hadoop & On-Prem Ecosystems

Hadoop ecosystem concepts, including HDFS, YARN, Hive, and Kerberos; experience with distributed storage/compute patterns, legacy-to-modern platform evolution, and secure access-controlled data environments.

Data Modeling & Tooling

Dimensional modeling, semantic modeling, schema design, metadata management, transformation layers, dbt concepts, data contracts, lineage, and modeling standards for analytics enablement and platform consistency.

Software Engineering, DevOps & Reliability

Git/GitHub, GitHub Actions, GitLab CI, CI/CD pipelines, automated testing, deployment automation, monitoring, logging, alerting, observability, incident response, idempotency, retries, SLA/SLO thinking, and fault-tolerant design.

Machine Learning & Data Pipelines

ML pipelines, MLOps concepts, feature engineering, data preparation, personalization workflows, LLM APIs, embedding pipelines, RAG concepts, and AI-enabled data workflows in production environments.

Data Security, Governance & Quality

Data privacy (PII), compliance-aware pipelines, access control, validation strategies, auditability, data quality checks, governance practices, and secure data lifecycle management.

Processes & Collaboration

High-ownership engineering mindset; Agile/Scrum; cross-functional collaboration with analysts, data scientists, engineers, product, infrastructure, and security teams; stakeholder enablement and translating complex requirements into scalable platform capabilities.